

Software Tools from Different Brands

Operating systems	Programming tools	Development tools
RTOS including kernels, Linux, Windows	Editors, compilers, assemblers, debuggers, libraries	Requirement analysis, Models, Simulators, UML, SysML, Testing, Co-design and verifications
Green Hill Software, Wind River, Wavecom, Express Logic, QNX, Mentor Graphics, Red Hat, Times, Microsoft, Enea, Segger, Kili	IAR Systems, AdaCore, Tasking, ACE, Python, Keil, Raisance, Macraigor, Mathworks, iSystem, GAIO Technology, Cosmic Software, Nokia-QT, ARM RVDS, Lauterbach	IBM Rational, MatrixOne, MKS, Altova, Telelogic, AWR, Altia, Rapita Systems, Virtio, Simpod, EVE, CoWare

Note: 1. This is not the complete list

2. Please check the website to know the processor/architecture supported by the tools

IDE in the form of graphical user interface (GUI) is mostly paired with the vendor's proprietary RTOS. For example—Wind River Workbench offers a collection of tools for software design, development, debugging, test and management through a tight integration with its RTOS VxWorks and Wind River Linux. "However every vendor may have different core competencies, some may offer very good debuggers, while others may offer easy to use editor. So, selection of IDE or using individual best tool set becomes very critical for a successful completion of project with respect to time and money," expresses Gurunatham.

Testing and bug fixing

With every new feature, designers at every stage of the embedded development cycle face more complex chal-

lenges. As the technology of hardware and software has evolved, system test and debug tools have also gone through a transition. To manage complexity, "visualising what you want to do in UML enables you to collaborate and create consistent designs," shares John. "With model-based approach, teams can visualise test cases for better communication with domain experts."

The best time to detect and fix a bug is early in the developmental process. The increasing complexity in systems has led to a new trend called "unit testing." A unit is a part of the design that can be tested individually. "Unit testing is very important in ensuring timing correctness. It is the first point at which you can see production code running on the target," says Almstrom. "Fixing timing problems at this

point means that cost of resolving the problem is much cheaper than leaving the problem for later stages."

After unit testing comes a more systematic technique called integration testing. It is used for system-level testing and detecting the errors associated with interfacing. UML models and analysis when combined with coherent integration testing can make developing quality embedded systems easier.

The test cases can be used during the entire development process to perform unit testing, integration and system testing and regression testing, using a common language. Software tools allow you to automate the test process, create test architectures, execute tests and automatically determine the success and failure of the test. The test results can be analysed and fixed to ensure the quality.

Multi-platform tools

The embedded development environment is extremely complex, you don't deal with just a single hardware platform or operating system from development to deployment. As the selection of chip or hardware has become cumbersome with so many options available today, software tools that are independent of the host (where the code is written) and the target (where the code runs) have increasingly become popular among developers.

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